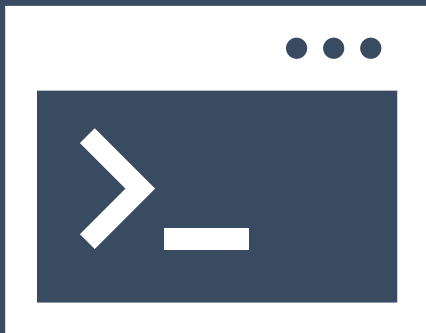


Investigating chromatin biology using ChIP-seq and CUT&RUN

<https://tinyurl.com/hbc-chromatin>



Harvard Chan Bioinformatics Core
in collaboration with
HMS Research Computing



Introductions!





Shannan Ho Sui
Director



Meeta Mistry
Associate Director



Lorena Pantano
*Director of Bioinformatics
Platform*



John Quackenbush
Faculty Advisor



Upen Bhattarai



Heather Wick



Will Gammerdinger



Noor Sohail



Alex Bartlett



Elizabeth
Partan



Emma Berdan



James Billingsley



Zhu Zhuo



Maria Simoneau



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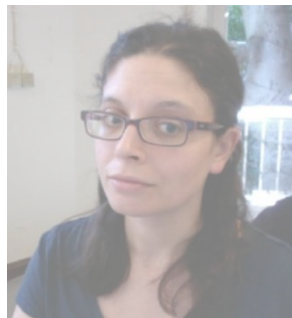
Noor Sohail



Alex Bartlett



Elizabeth
Partan



Emma Berdan



James Billingsley



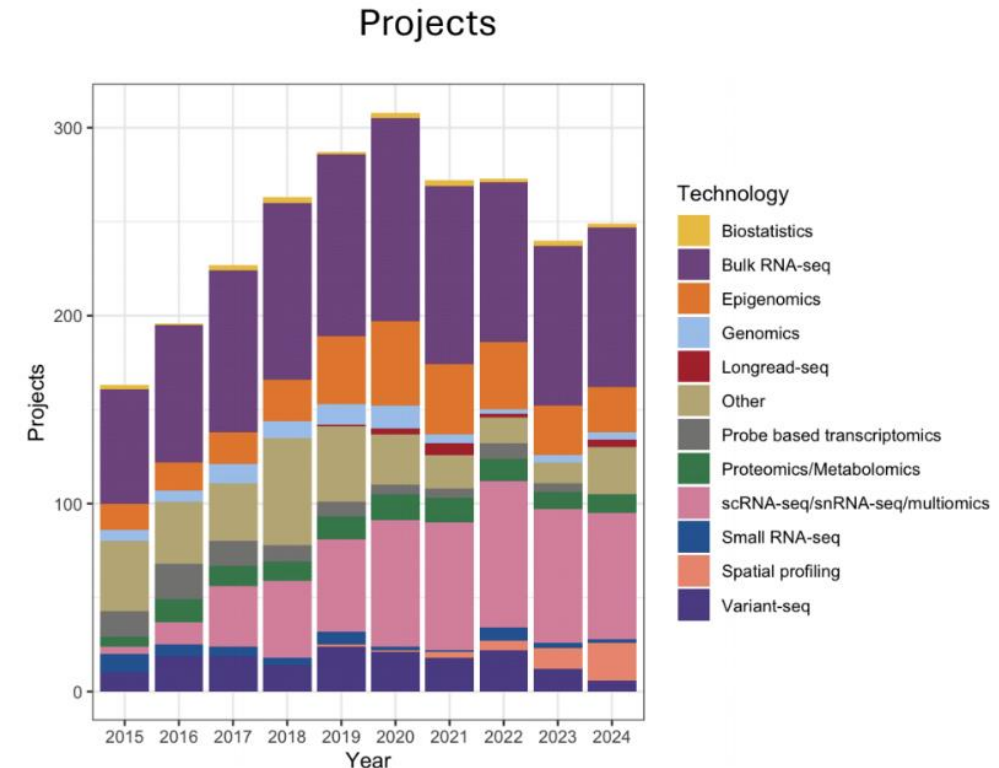
Zhu Zhuo



Maria Simoneau

Consulting

- ❖ Transcriptomics: Bulk, single cell, small RNA
- ❖ Epigenomics: ChIP-seq, CUT&RUN, ATAC-seq, DNA methylation
- ❖ Variant discovery: WGS, resequencing, exome-seq and CNV
- ❖ Multiomics integration
- ❖ Spatial biology
- ❖ Experimental design and grant support



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SCIENCE CENTER



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MEDICAL SCHOOL

Training

- ❖ Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design

- ❖ Basic Data Skills

- ❖ Shell

- ❖ R

- ❖ Advanced Topics: Analysis of high-throughput sequencing data

- ❖ Chromatin Biology

- ❖ Bulk RNA-seq

- ❖ Differential Gene Expression

- ❖ scRNA-seq

- ❖ Variant Calling

- ❖ Current Topics in Bioinformatics

<https://bioinformatics.sph.harvard.edu/training>

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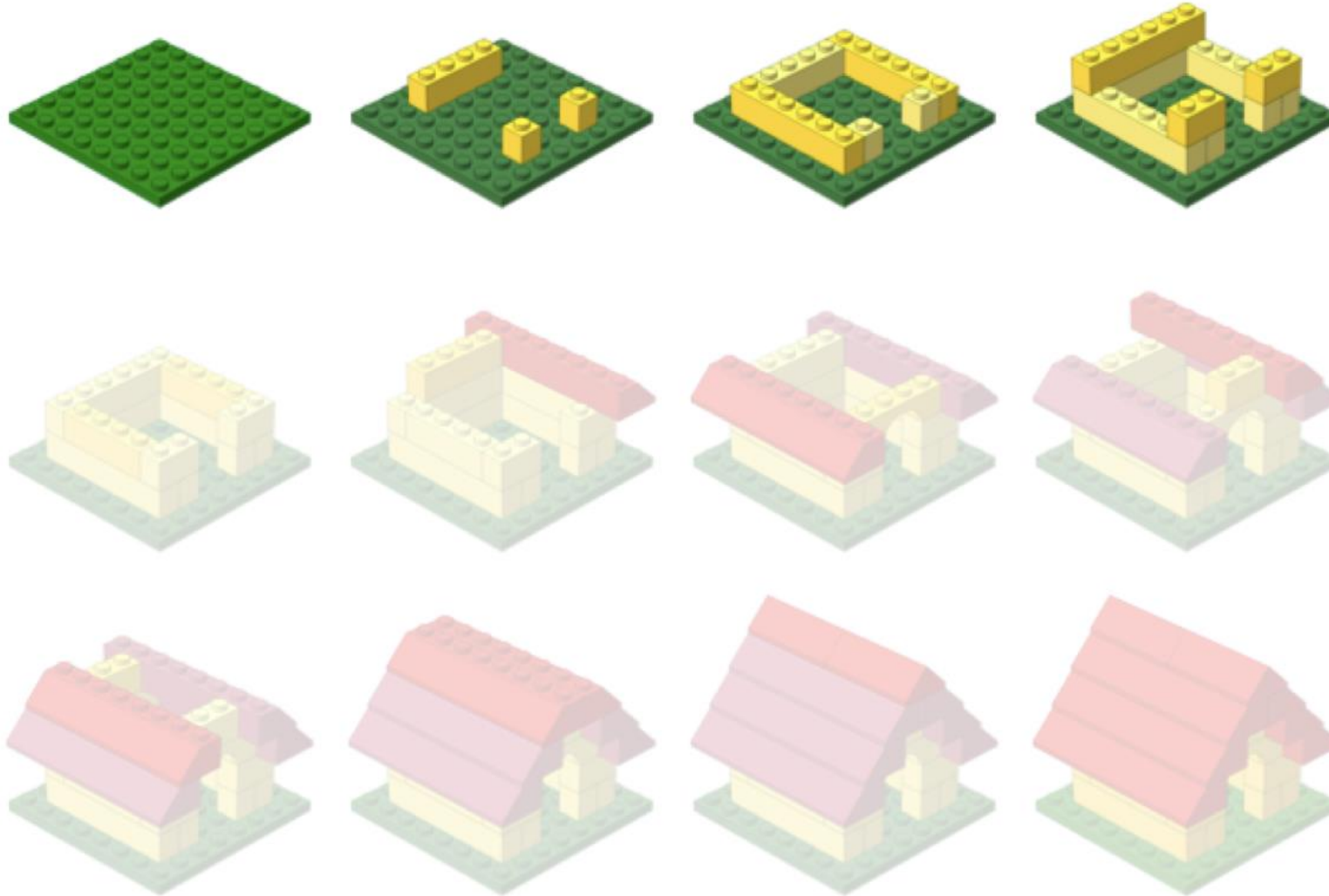
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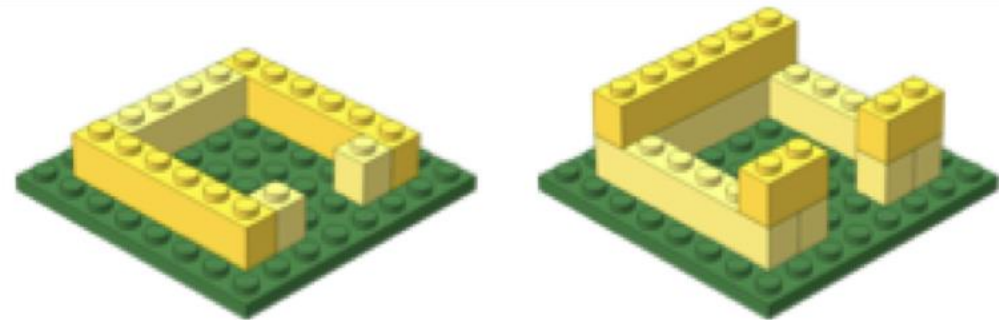
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Setting up to perform Bioinformatics analysis

Setting up...



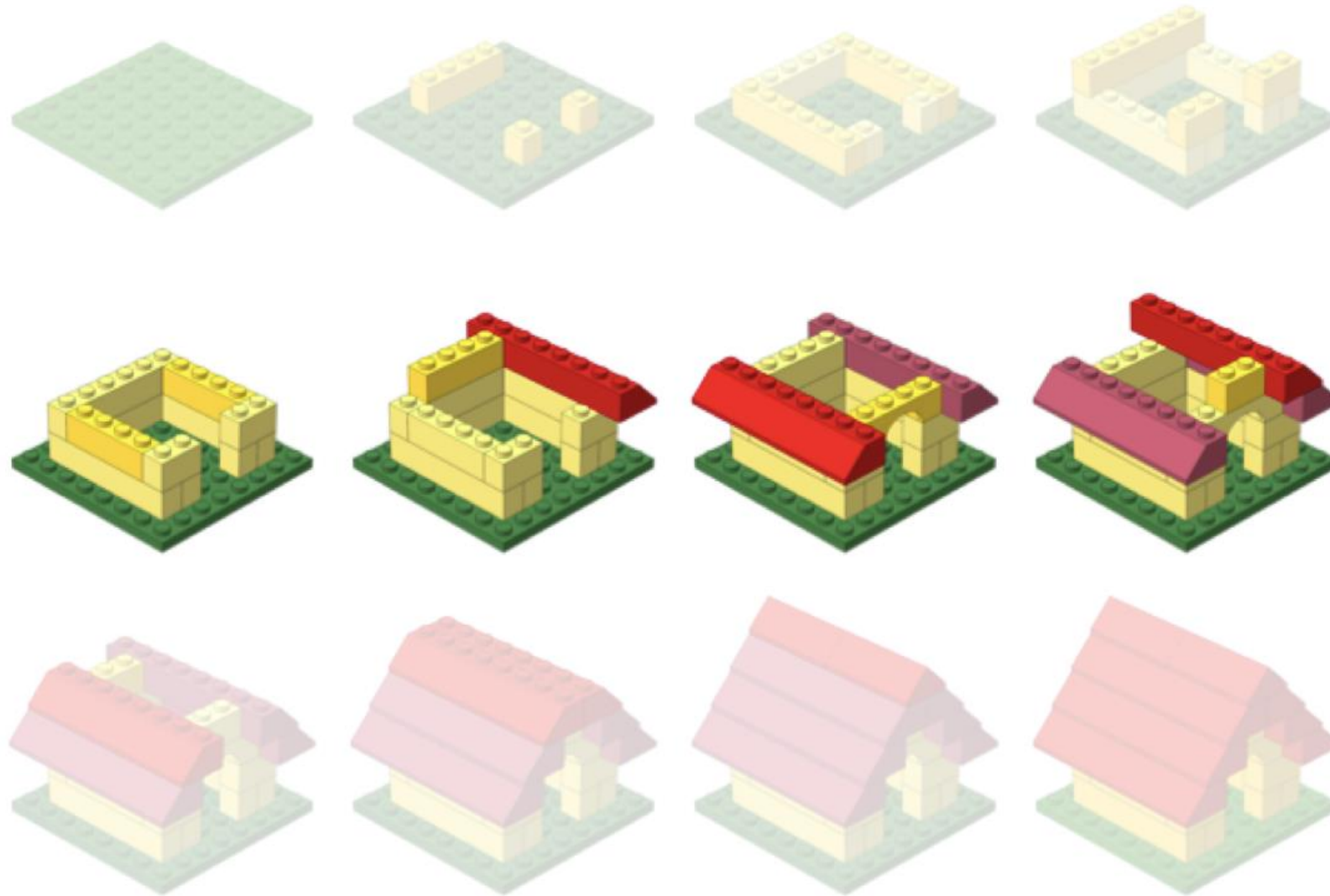
❖ Shell for Bioinformatics

- ❖ Dealing with large data files
- ❖ Performing bioinformatics analysis
- ❖ Using tools
- ❖ Accessing and using compute clusters

❖ Introduction to R

- ❖ Parsing and working with smaller BED files
- ❖ Statistical analysis, e.g. differential binding analysis
- ❖ Generating figures from complex data

Workshop scope



Bioinformatic Data Analysis

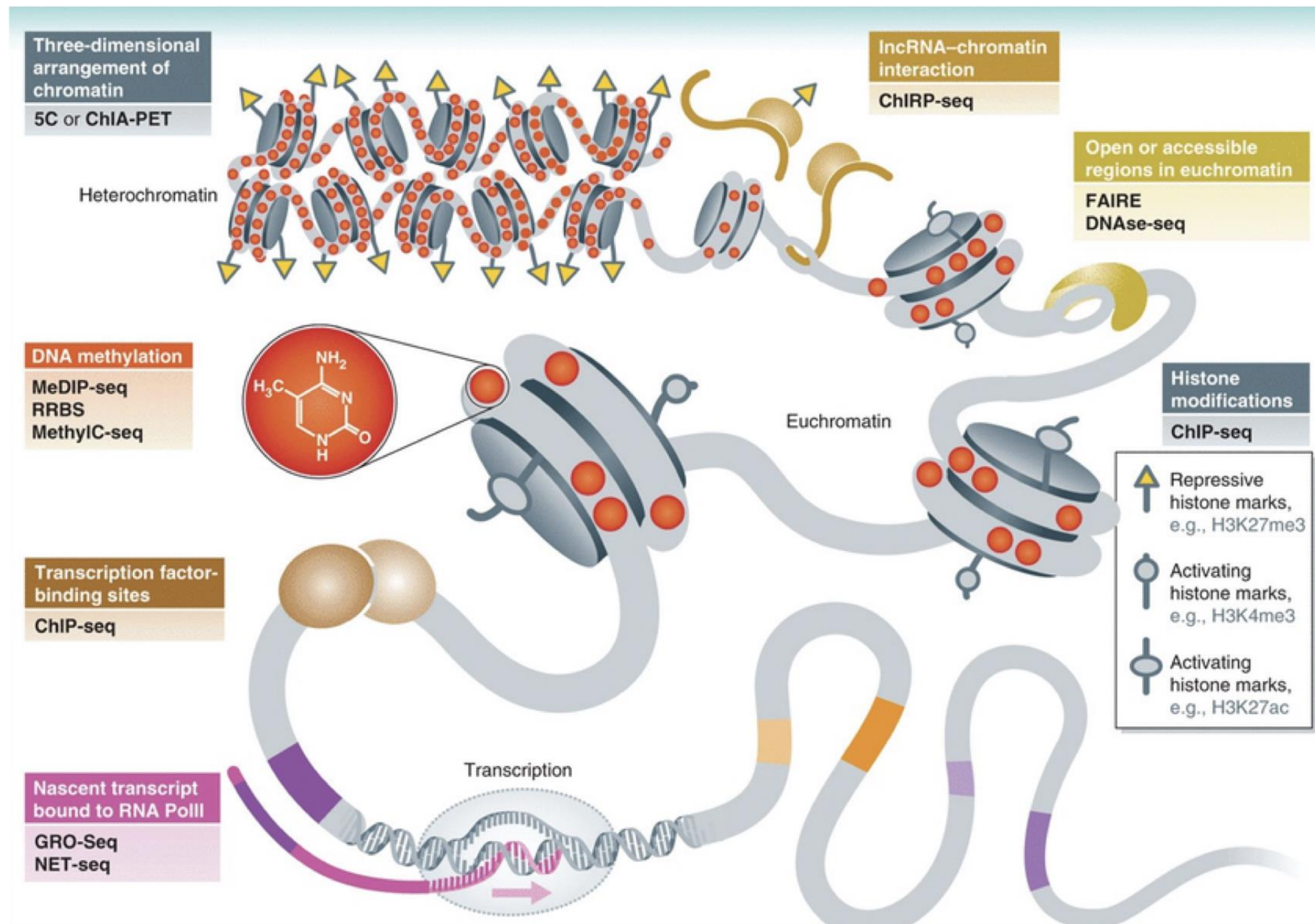
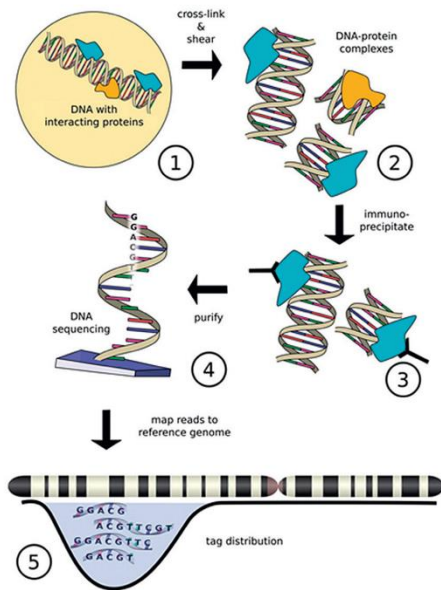


Figure adapted from Soon WW, Hariharan M, Snyder MP, “High throughput sequencing for biology and medicine”. Molecular Systems Biology 9:640 2013

Genomic Methods for Profiling Chromatin

ChIP-seq



CUT&RUN

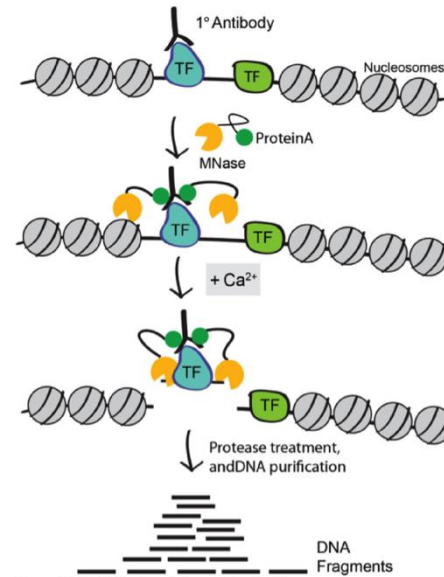
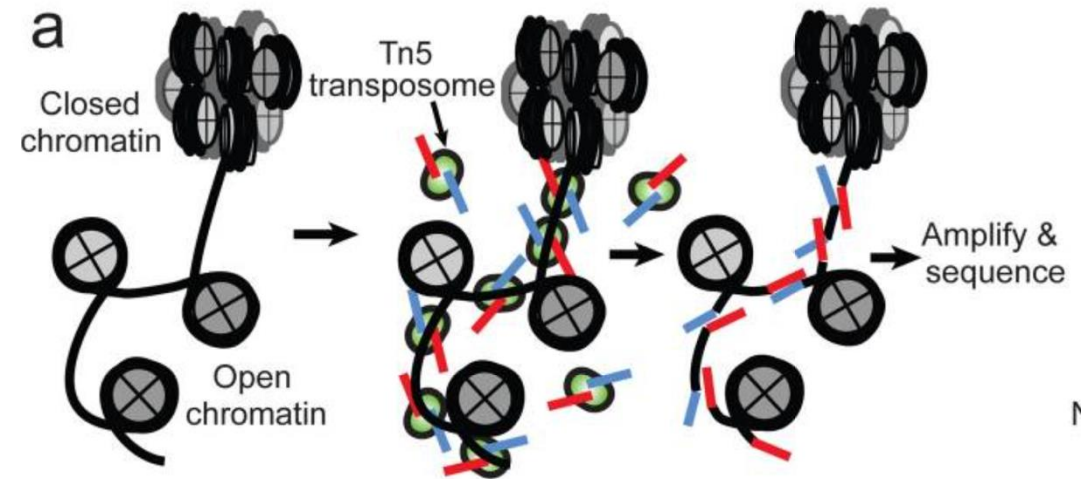
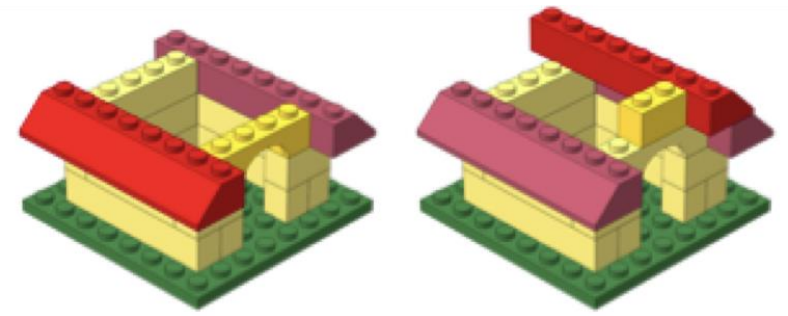


Figure 1. CUT&RUN schematic (see text for details).

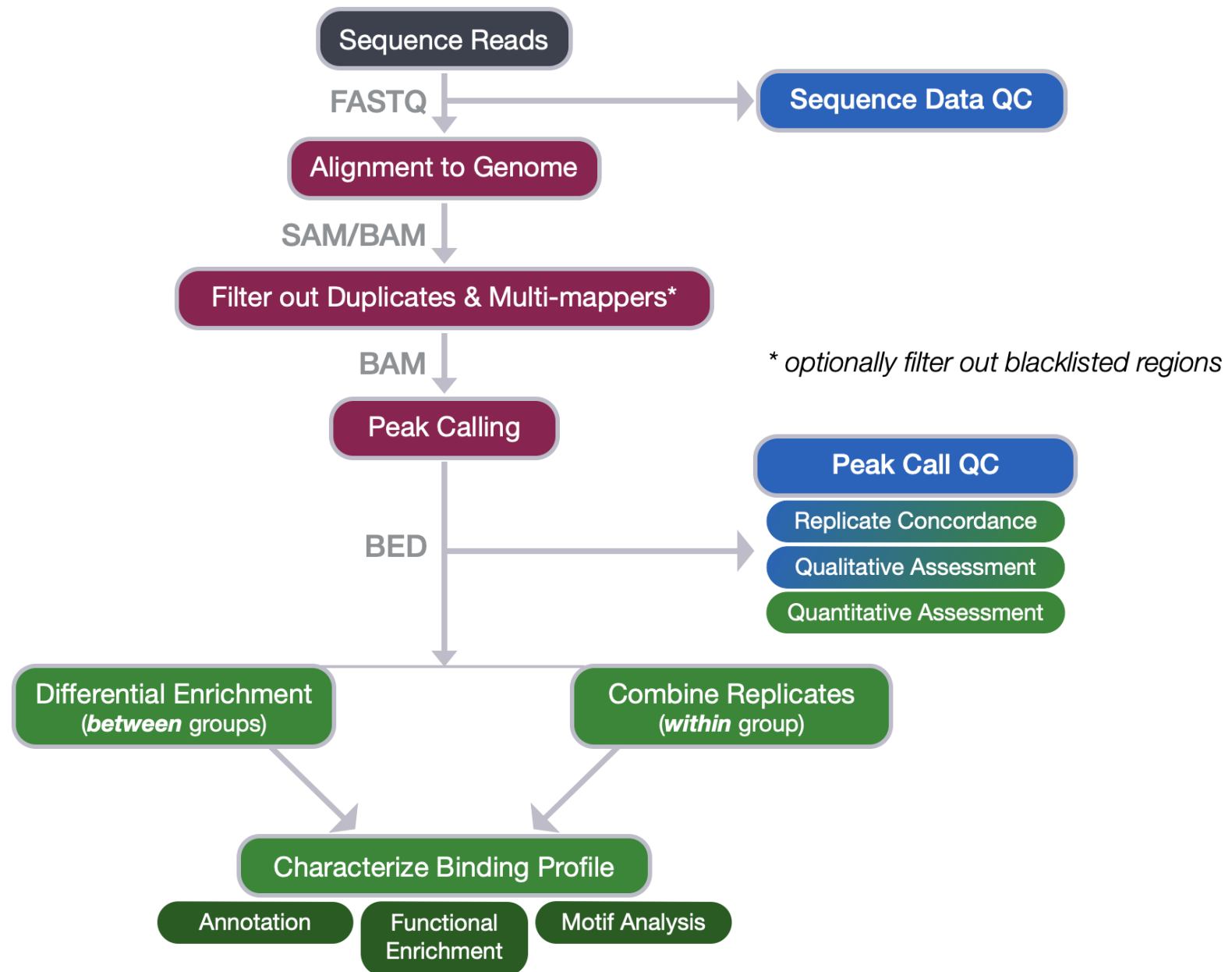
ATAC-seq



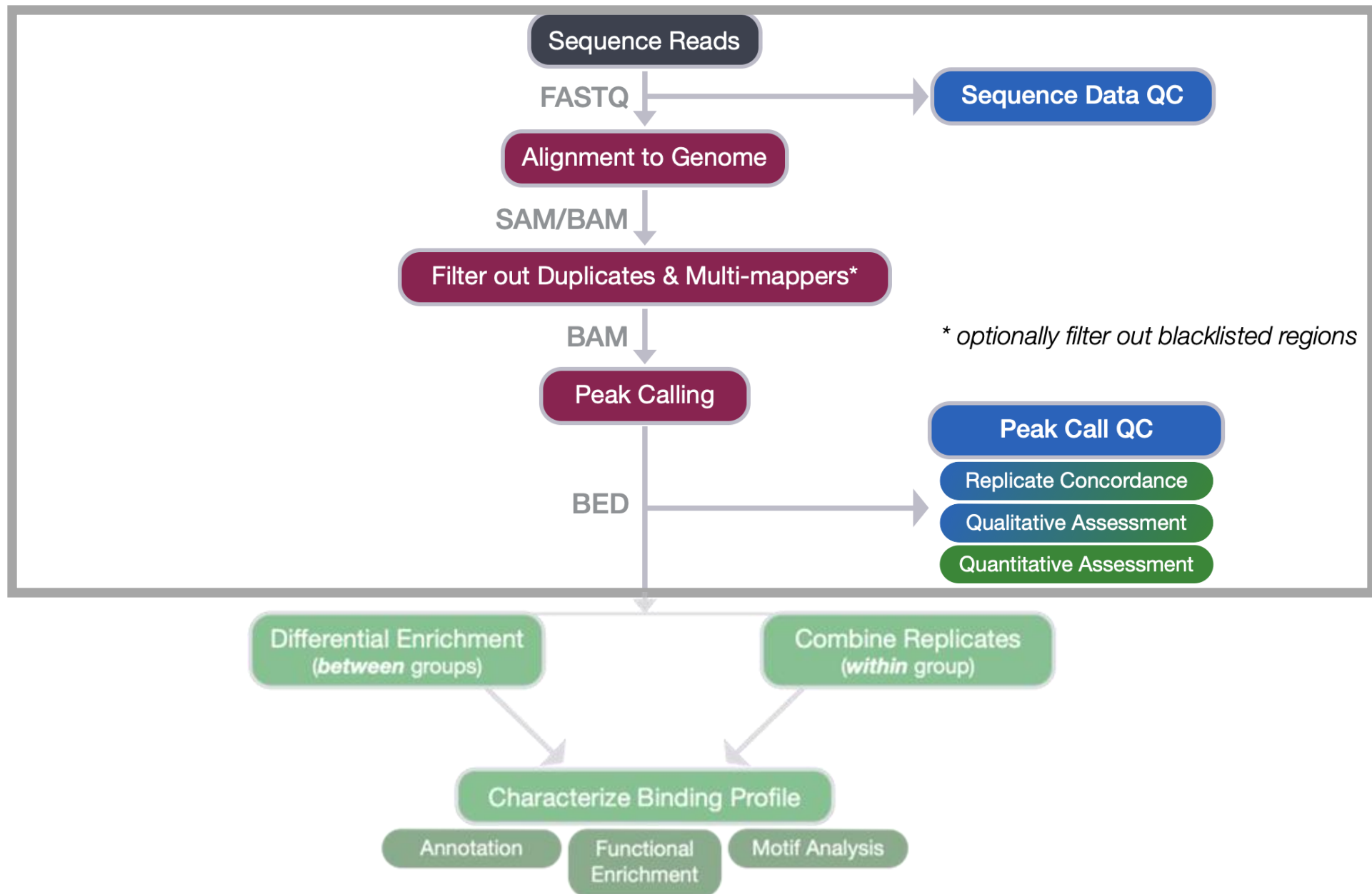
Learning Objectives



- ❖ Describe important considerations for setting up a successful ChIP-seq, CUT&RUN or ATAC-seq experiment
- ❖ Describe the steps in a ChIP-seq analysis workflow (from sequence data to peak calls) and contrast any differences for CUT&RUN and ATAC-seq analyses
- ❖ Learn how to handle various file formats encountered when analyzing ChIP-seq and related data
- ❖ Implement shell scripts on a high-performance compute cluster to perform the above steps

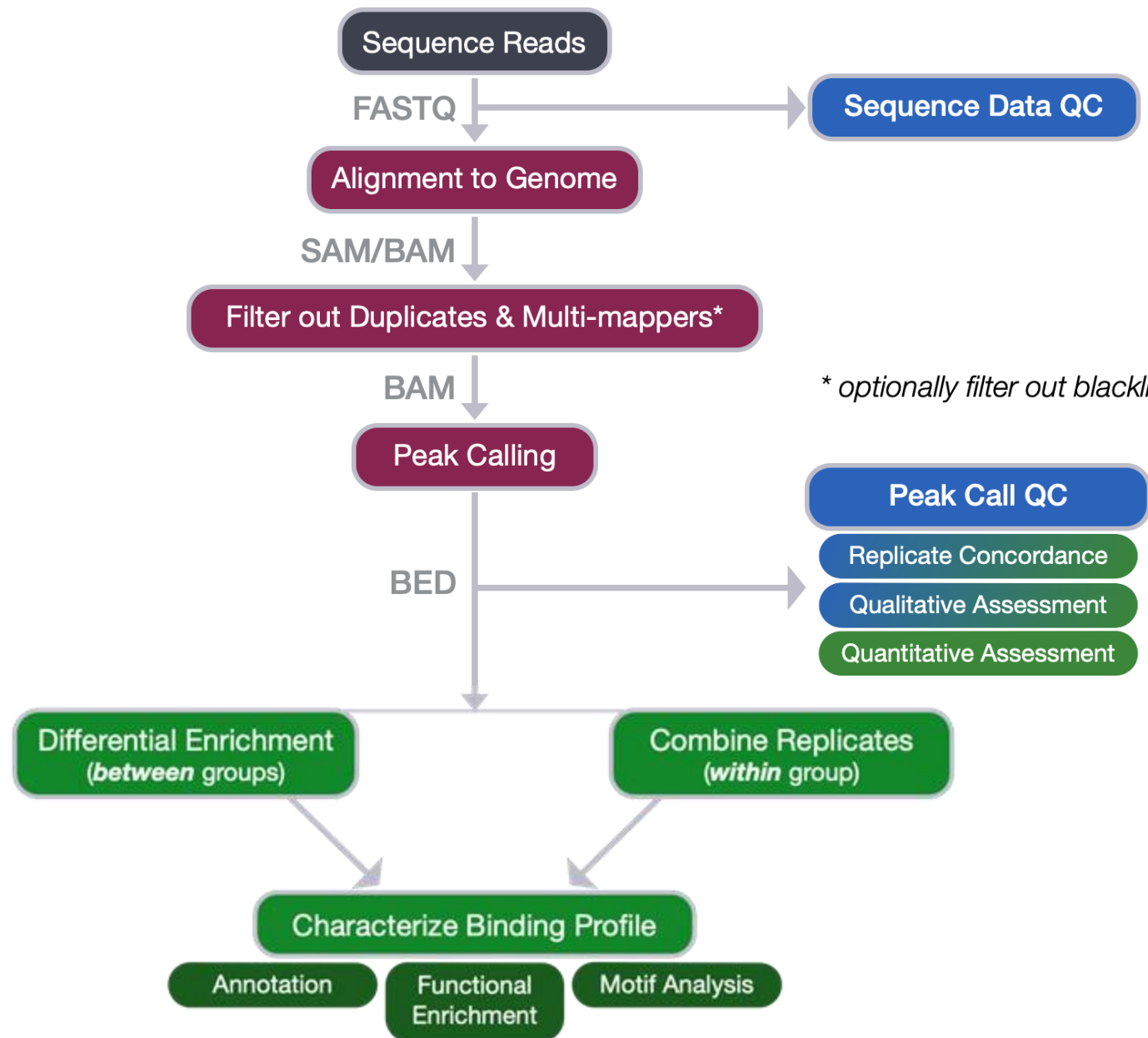


The Workflow



The Workflow

Boxes in green represent parts of the workflow that will not be covered in this workshop



The Workflow

Boxes in green represent parts of the workflow that will not be covered in this workshop

Logistics



Course schedule

Day 1

Time	Topic	Instructor
09:30 - 09:45	Workshop Introduction	Meeta
09:45 - 11:00	Studying epigenetic regulation using high-throughput sequencing data	Dr. Ming "Tommy" Tang
11:00- 11:05	Break	
11:05 - 11:20	HPC review Q&A	Will
11:20 - 11:50	Dataset overview and project organization	Will
11:50 - 12:00	Overview of self-learning materials and homework submission	Meeta

Before the next class:

I. Please **study the contents** and **work through all the code** within the following lessons:

1. [Experimental design considerations for HTS of chromatin](#)

[Click here for a preview of this lesson](#)

<https://tinyurl.com/hbc-chromatin>

Course materials

- ❖ We continuously update our materials to reflect changes in the field/software



Approximate time: 45 minutes

Learning Objectives

- Learn about quality control considerations when preparing your ChIP-seq experiment
- Understand the need for positive and negative controls

<https://tinyurl.com/hbc-chromatin>

Course participation

- ❖ Mandatory review of self-learning lessons and assignments
- ❖ Attendance required for all classes
- ❖ Your questions and active participation drive learning
- ❖ **We look forward to all of your questions!**



Course participation

- ❖ At-home lessons and exercises after each session
- ❖ Cover material not previously discussed
- ❖ Provides us feedback to help pace the course appropriately
- ❖ 3-5 hours to complete
- ❖ Homework load is heavier in the beginning of this workshop series and tapers off

Using AI for Assignments

❖ Do

- ❖ Try to resolve error messages with it
- ❖ Test code written by AI on a dataset where you have expected results
- ❖ Take the time to review the generated code line-by-line

❖ Don't

- ❖ Implement it in replacement to learning
- ❖ Write code that you don't understand
- ❖ Assume the output from an AI process is correct

Odds & Ends

- ❖ Quit/minimize all applications that are not required for class
- ❖ Name tags
- ❖ Post-its
 - ❖  green - I am all set
 - ❖  red - I need time/help
- ❖ Phones on vibrate/silent
- ❖ Bathrooms

Thanks!

- ❖ Kathleen Chappell and Andy Bergman from HMS-RC
- ❖ Data Carpentry

These materials have been developed by members of the teaching team at the Harvard Chan Bioinformatics Core (HBC). These are open access materials distributed under the terms of the Creative Commons Attribution license (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Contact Us

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- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu
- ❖ *O2 (HMS-RC):* rchelp@hms.harvard.edu